

Ryan Chen

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EDUCATION

Stony Brook University

Bachelor of Science in Computer Science, GPA: 3.67/4.00

- **Honors:** Dean's List (Fall 2025, Spring 2026)
- **Relevant Coursework:** Data Structures, Discrete Mathematics, Principles of Programming Languages, Systems Fundamentals (C, Assembly)

Stony Brook, NY

Expected May 2029

TECHNICAL SKILLS & CERTIFICATIONS

Languages: Java, Python, C, TypeScript, JavaScript, Kotlin, Swift, SQL, HTML/CSS, Assembly

Frameworks & Libraries: React.js, Next.js, Material-UI

Developer Tools: Git, PostgreSQL, Supabase, Vercel, VS Code, PyCharm, IntelliJ, Android Studio, Xcode, Bash

Certifications: Google Cybersecurity (Coursera), Meta Front-End Developer (Coursera)

EXPERIENCE

Headstarter AI

Software Engineering Fellow

- Engineered AI-driven projects including Pantry Tracker, AI flashcards, and an AI customer support system using Next.js, Material-UI, and the OpenAI and Gemini APIs

New York, NY

Jul 2024 – Sept 2024

NYCDOE Division of Instructional and Information Technology

Intern

- Presented key insights to CIO and executive team, influencing strategic decisions
- Analyzed TeachHub usage and surveyed 500+ K-12 students to improve functionality for the largest U.S. school district (912,064 students)

Brooklyn, NY

Jul 2024 – Aug 2024

Project Intern

- Standardized official transcript processes across NYC DOE systems by designing detailed UML diagrams and workflow proposals
- Documented edge cases and created scenarios to ensure accurate record delivery across departments
- Examined student records including transcripts, IEPs, 504s, and PSAL documentation to inform recommendations

Apr 2024 – May 2024

INDEPENDENT PROJECTS

Bao-bae – Live Audience Voting App | *Next.js, React 19, TypeScript, Supabase, PostgreSQL*

April 2026

- Co-developed and deployed mobile-first voting platform for a live campus show, processing 219 votes across 3 elimination rounds with Google OAuth authentication
- Enforced one-vote-per-round integrity via composite unique constraints and Row-Level Security policies, converting PostgreSQL constraint violations into user-facing error states
- Delivered near-real-time results by polling a PostgreSQL aggregate view at 1.5s intervals, avoiding websocket infrastructure while supporting live vote totals

Pharmacy Refill Database (Contract) | *PostgreSQL, SQL, Bash, Twilio API*

October 2025

- Owned the data layer on a remote team building a prescription refill platform, designing a normalized 6-table PostgreSQL schema modeling the full refill lifecycle from request through pickup
- Enforced referential integrity across 5 foreign-key relationships and prevented duplicate prescriptions per patient with a composite unique constraint, backing status and request-source fields with lookup tables
- Optimized read paths with composite indexes on patient-prescription and name-DOB lookups, exposing a denormalized view joining patients, refills, and status for application queries

Nue-Trivia (Hackathon Winner) | *Java, OpenAI API, JFrame, GPT-4*

October 2024

- Designed educational nutrition game featuring custom sprites and AI-generated questions for young audiences
- Constructed interactive Q&A gameplay with point-based reward system to enhance child learning engagement

Pantry Tracker | *JavaScript, React.js, Material-UI, Gemini API*

August 2024

- Developed responsive web app to track pantry inventory and provide AI-powered recipe suggestions
- Implemented dynamic search and modals for inventory management with fully mobile-optimized UI